

Market Power and Contagion in Interbank Networks

An Agent-Based Model of Systemic Risk

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Outline

- 1 Abstract
- 2 Risk and Connectivity
- 3 Model Framework
- 4 Interest Rate and Market Power
- 5 Equity and Bankruptcy Dynamics
- 6 Phase Transition
- 7 Results
- 8 Conclusions

Abstract: ABM, Interbank Market, and Contagion

- **Agent-Based Models** to study systemic risk in interbank credit networks.
- We examine the relationship between **network connectivity**, **market power**, and the propagation of **bankruptcy cascades**.
- **Non-monotonic trade-off** between risk sharing and systemic risk—connectivity diversifies risk in normal times but amplifies contagion during crises.
- The 2008 financial crisis illustrates how risk-sharing mechanisms can fail under high leverage and heterogeneous agent populations.

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The 2008 Crisis: Background

- Increased bank leverage and corporate debt-equity ratios made the economy fragile.
- Standard competitive equilibrium models do not capture the complex interactions that amplify crises.
- The interbank market became a **contagion vector**: failure of one bank triggers a domino effect through interlocking exposures.

The Risk-Sharing Trade-off

Two opposing effects of network connectivity:

- **Benefit:** Increasing connectivity allows **risk sharing**—banks diversify exposure across more counterparties.
- **Cost:** Excessive connectivity increases **systemic risk**—default propagates through the network.
- Evidence: a **non-monotonic relationship** exists—low connectivity stabilizes by diluting shocks, but high connectivity facilitates cascading failures.

Channels of Contagion in Financial Crises

Three primary propagation mechanisms of systemic failure:

- 1 **Bank runs:** self-fulfilling panics where depositors withdraw mass funds simultaneously.
- 2 **Asset price contagion:** fire-sale spirals that depreciate assets across the system.
- 3 **Interlocking exposures:** direct linkages through the interbank market—default of one bank triggers defaults in connected banks.

The model distinguishes three failure channels: **rationing** (borrower cannot secure sufficient funding), **contagion** (lender suffers losses from borrower default), and **repayment failure**.

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Model Overview

- ABM with N banks in a sequential economy $t = \{0, 1, \dots, T\}$.
- Banks connected through an **Erdős–Rényi** random graph $G(N, p)$: each edge exists with probability p .
- Three interacting sectors: goods market, credit market, interbank market.
- Goal: analyze how **connectivity** and **market power** affect system stability.

Balance Sheet

Accounting Identity

$$L_i^t + C_i^t + R_i^t = D_i^t + E_i^t$$

- **Assets:** Long-term loans (L), liquidity (C), reserves (R).
- **Liabilities:** Deposits (D), equity (E).
- Reserves set by regulation: $R_i^t = \hat{r}D_i^t$ with $\hat{r} = 0.02$.

Liquidity Shocks

Deposits are exogenously shocked each period:

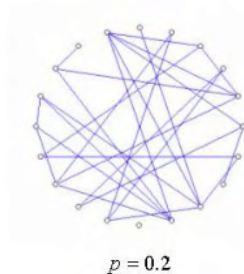
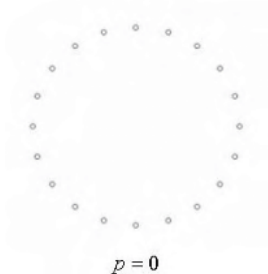
$$D_i^t = D_i^{t-1} [\mu + \omega \cdot U(0, 1)]$$

- μ : expected growth rate (0.7).
- ω : shock magnitude (0.35).
- **Potential lenders:** banks with positive net deposits (after reserves).
- **Potential borrowers:** banks with negative net deposits exceeding liquidity.

Network Structure

- Bilateral credit lines formed at the start of each period.
- Connectivity controlled by parameter p in the Erdős–Rényi graph.
- Each borrower is matched to **exactly one** lender per period.
- The **Giant Component Size (GCS)** measures network integration.

Erdős–Rényi Random Graph



Each of the $\binom{N}{2}$ possible edges is included independently with probability p . For $N = 50$ banks:

- $p = 0$: isolated banks, no interbank market.
- $p = 1/N = 0.02$: **percolation threshold**—giant component emerges.
- $p > 0.05$: network is saturated ($> 92\%$ of banks connected).

Network Structure

The Erdős–Rényi model does not attempt to replicate the entire interbank market; rather, it illustrates how connection density alters the probability and scale of contagion.

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Expected Profit

Banks are risk-neutral in perfect competition. Expected profit from lending to borrower j :

$$E[\Pi_{i,j}^t] = p_j^t(r_{i,j}^t - m)c_{i,j}^t - (1 - p_j^t)m \cdot c_{i,j}^t$$

- m : screening cost (0.015).
- p_j^t : probability of borrower survival.
- $r_{i,j}^t$: interest rate; $c_{i,j}^t$: lending capacity.

Interest Rate Derivation

Setting $E[\Pi] = 0$ (zero-profit condition):

$$r_{i,j}^t = \frac{m}{p_j^t}$$

Introducing **market power** $\psi \in [0, 1)$:

$$r_{i,j}^t = \frac{m}{p_j^t(1 - \psi)}$$

- $\psi = 0$: competitive (actuarially fair) rate.
- $\psi > 0$: monopolistic markup from information asymmetry.

Endogenous Market Power

Market power evolves endogenously with bank size:

$$\psi_i^t = \frac{E_i^t}{E_i^{t,\max}}$$

- Larger banks (higher equity) exercise more market power.
- Borrower default probability: $p_{b,j}^t = E_j^t / E_j^{t,\max}$.
- Leverage: $\lambda_j^t = d_j^t / E_j^t$.
- Within the model, market power and default probabilities are determined **endogenously** from screening costs and shock parameters.

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Equity Evolution

$$E_i^t = E_i^{t-1} + \sum_j l_{i,j}^t r_{i,j}^t - \sum_{j \subseteq \theta_i^t} [B_{i,j}^t - \tilde{L}_j^t]$$

- **Bankruptcy:** bank exits if $E_i^t < 0$.
- **Fire sales:** last-resort option (blocked in this model to amplify contagion effects).
- Failed banks replaced by new entrants with small initial capital.

Failure Channels

Three distinct causes of bank failure:

- ① **Rationing:** borrower cannot secure sufficient funding.
- ② **Contagion:** lender suffers losses from borrower default (bad debt).
- ③ **Repayment failure:** fully-funded borrower cannot repay loan or interest.

Repayment failure dominates at high connectivity, while rationing dominates at low connectivity.

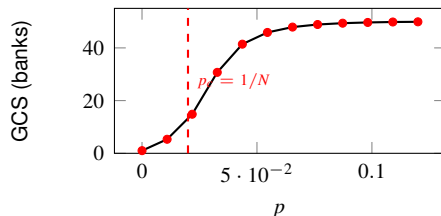
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Phase Transition

- **Percolation threshold:** GCS jumps from ~ 1 to ~ 46 banks (92%).
- **Low p :** rationing-dominated, high interest rates, few loans.
- **High p :** repayment-failure-dominated, low rates, saturated network.
- $p > 0.05$: all macroeconomic metrics stabilize.

Phase Transition



Percolation threshold for

Erdős–Rényi graph:

$$p_c = \frac{1}{N} = \frac{1}{50} = 0.02$$

p	GCS
0.00001	1
0.011	5
0.022	15
0.033	31
0.044	41
0.055	46
0.065	48
0.087	49
≥ 0.11	50

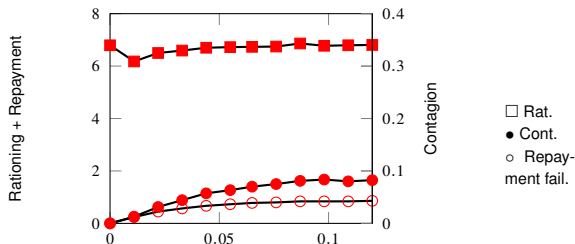
Connectivity and Systemic Risk

- As p increases, the interest rate drops from 30 (upper bound) to ~ 1.65 .
- Bankruptcies remain stable at ~ 7 – 7.5 (limited by model structure).
- Active loans saturate at ~ 13 (26% of banks).
- Equity jumps from 46.8 to ~ 55 and then plateaus.
- Bad debt grows rapidly before stabilizing at ~ 1.3 .

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Bankruptcies exposed by the model



p	Rat.	Cont.	Fail.
0.001	6.79	0.00	0.00
0.011	6.18	0.01	0.24
0.022	6.50	0.03	0.45
0.033	6.59	0.04	0.57
0.044	6.70	0.06	0.67
0.055	6.72	0.06	0.73
0.065	6.73	0.07	0.78
0.076	6.75	0.07	0.80
0.087	6.86	0.08	0.84
0.098	6.77	0.08	0.84
0.109	6.79	0.08	0.84
0.120	6.81	0.08	0.86

Assets

Liabilities

Initial conditions for each bank:

$$C_0^i = 5 - 0.18 = 4.82$$

$$L_0^i = 5$$

$$R_0^i = 0.18$$

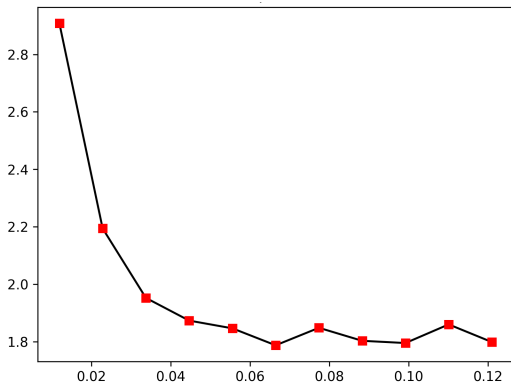
$$D_0^i = 9$$

$$E_0^i = 1$$

Monte Carlo simulations with $\mathcal{MC} = 50$ runs, $N = 50$ banks, $T = 1000$ periods, and p varying from 0 to 0.12 in increments of 0.01.

Interest rate

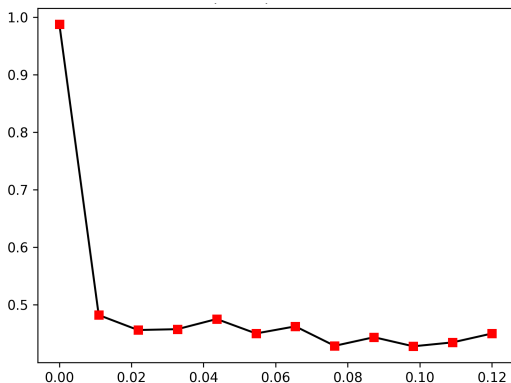
r^i is decreasing in p because higher connectivity reduces market power



p	r^i
0.012	2.9083
0.023	2.1940
0.034	1.9521
0.045	1.8731
0.056	1.8461
0.066	1.7871
0.077	1.8480
0.088	1.8027
0.099	1.7949
0.110	1.8598
0.121	1.7982

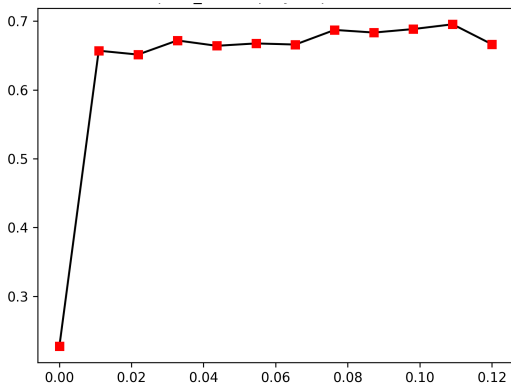
Market power ψ

ψ decreases as connectivity p increases (more competition).



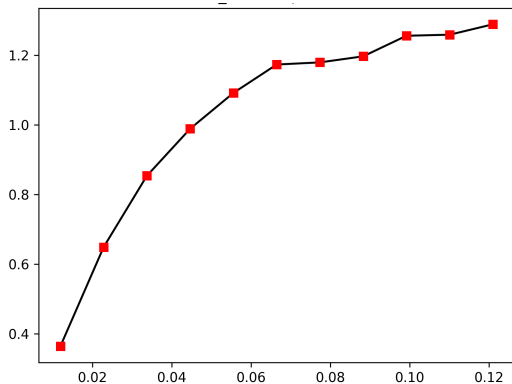
p	ψ
0.001	0.9875
0.011	0.4823
0.022	0.4563
0.033	0.4577
0.044	0.4754
0.055	0.4503
0.065	0.4626
0.076	0.4288
0.087	0.4438
0.098	0.4280
0.109	0.4348
0.120	0.4502

Probability of bankruptcy p_b



p	p_b
0.001	0.2273
0.011	0.6571
0.022	0.6514
0.033	0.6719
0.044	0.6642
0.055	0.6676
0.065	0.6660
0.076	0.6873
0.087	0.6834
0.098	0.6885
0.109	0.6954
0.120	0.6663

Bad debt



p	B
0.012	0.3636
0.023	0.6486
0.034	0.8536
0.045	0.9888
0.056	1.0917
0.066	1.1733
0.077	1.1794
0.088	1.1968
0.099	1.2559
0.110	1.2590
0.121	1.2888

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Key Findings

As connectivity p increases:

- 1 The Giant Component Size increases sharply above the percolation threshold.
- 2 Market power decreases.
- 3 The equilibrium probability of bankruptcy increases with connectivity.
- 4 The equilibrium interest rate falls.

Key Findings

- 1 **Failure composition shifts:** rationing dominates at low p ; repayment failure dominates at high p .

Policy Implications

- **Heterogeneity** is a root cause of systemic instability; the failure of large banks has disproportionate systemic effects.
- Interbank lending should ideally be restricted among banks with **similar liquidity characteristics**.
- Reserve requirements dampen cascades but **constrain growth**.
- Risk diversification across many agents has **ambiguous** effects on total systemic stability.